Generation Recovery and Validation

This handbook describes the supported operator path for recovering local image and video generation after an interrupted install. It applies to the packaged Windows installer and installed Nexus desktop application.

Saved Image Studio edits have a separate recovery path. Each agent approval attempt captures the complete saved edit and undo/redo history before access can be granted; a storage failure grants no access. In the image viewer, open "Recovery checkpoints", inspect the approval metadata, choose a checkpoint and confirm "Restore and save". Pending edits are saved first. Recovery replaces the current edit history, ends earlier agent access to that image and refuses a conflicting revision; reload if another editor changed it. Deleting an image session or folder removes its history, edits and checkpoints, while shared original image files stay on disk.

Recover the runtime

1. Close Nexus and any active installer window.
2. Rerun the exact `NexusSetup.exe` candidate to provision or recover the local runtime.
3. Let the installer reclaim only a lease whose owner is proven dead or mismatched. A live or uncertain owner remains protected.
4. Treat the install as successful only when the runtime readiness probe reaches `ready`. A selected media capability that remains unavailable is a required failure.

Validate image and video generation

Install the exact candidate first and retain evidence that the installed payload came from it. Then run `scripts/installer/build/smoke-installed-media.ps1` against that installer artifact and installed sidecar. The harness records the supplied installer hash and exercises the installed runtime; it does not install the candidate or establish that binding itself. Retain one decodable PNG and one playable video, runtime versions, model ids, elapsed time, artifact hashes, and probe metadata.

The PNG proof includes format, dimensions, non-zero pixel content, and SHA-256. The video proof includes codec, duration, frame count, dimensions, and SHA-256. Unit tests and mock CUDA runs do not replace this packaged evidence.

Interpret failures

- `REPAIR_BUSY` means a verified live owner holds the repair lease.
- `REPAIR_OWNER_UNKNOWN` means ownership cannot be proved safely; stop the competing process or reboot before retrying.

- `runtime-unavailable` means the installed media environment failed readiness. Repair is available when its required components are present; missing components require rerunning the installer. Generation remains unavailable until readiness succeeds.
- A missing PNG or video after the exact packaged run blocks release publication.

See the installer runtime and media runtime companions for component ownership.